

DATA CLEAN AGENT

Autonomous AI-Powered Data Restoration & Audit Engine

HACKATHON TECHNICAL REPORT

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Version: 1.0 (Production)

Executive Summary:

Data preparation occupies up to 80% of a data scientist's time, and arbitrary AI-based cleaning is often rejected due to lack of auditability. DataCleanAgent solves this by combining autonomous LLM reasoning with a deterministic, sandboxed execution registry. The platform profiles datasets against 6 dimensions of quality, plans and executes cleaning steps using a ReAct reasoning model, verifies each change against schema-drift limits, and outputs both professional DOCX audit reports and reproducible Python code.

Technology Stack

- User Interface: Streamlit (custom dark-mode glassmorphic cyber-aesthetic theme)
- AI Agent Brain: Google Gemini 2.5 Flash (via structured JSON Outputs API)
- Database / Session Store: SQLite (tracks historical cleaning sessions, action logs, and data quality states)
- Data Visualization: Plotly (animated gauge indicators, dual-radar overlays, and column-level quality breakdown)
- Data Libraries: Pandas, NumPy, Scikit-learn, and Difflib (fuzzy label normalization)
- Report Export: python-docx (with custom professional typography and margins)
- QA Testing: Python Unittest (48 automated test cases)

MVP Scope vs. Premium Extras Added

Core MVP Deliverables

- 6-Dimension Profiler evaluating Completeness, Uniqueness, Validity, Consistency, Accuracy, and Integrity.
- ReAct Agent Loop framework using Gemini to diagnose and clean tabular files.
- Deterministic Execution Registry decoupled from dangerous raw `exec()` commands.
- Basic Audit Trails and historical logging.
- Auto-Generated DOCX Report detailing quality checks and methodology.

Premium Extras (Wow Factor Additions)

- Unified Data Health Score Gauge: Animated Plotly dashboard chart showing aggregated quality scores (0-100%) and before/after improvements.
- Reproducible Python Script Export: Generates clean, production-ready pandas scripts to replicate the AI's actions.
- Multi-Step Session Rollback: Reverts data changes and Gemini chat memory dynamically.
- LLM Self-Healing Engine: Feeds execution errors back to the model to automatically re-plan with correct arguments.
- Data Drift Guardrails: Monitors and alerts on unexpected row-loss, column drop, or structural changes.
- Pre-Loaded Datasets: UCI Adult Income, Dirty Data Challenge REIN, OpenRefine, and Synthetic Messy Titanic built-in for instant demo validation.

Detailed Feature List & Quality Metrics

Data Quality Operations Suite

- Median/Grouped Imputation: Auto-imputes missing values globally or based on subgroup averages.
- Fuzzy Category Normalization: Merges text typos and case inconsistencies (e.g. USA, U.S.A., usa -> USA).
- Outlier Clipping: Identifies and limits outliers using robust IQR and Z-Score thresholds.
- Range & Rule Clamping: Clips negative numeric fields to 0.0 and coerces invalid range records to Null.
- Encoding/Delimiter Ingestion: Automatically parses various CSV formats, BOMs, and separation tokens.

Security & AI Guardrails

To ensure corporate safety, the AI is completely decoupled from raw Python execution. It outputs structured JSON containing specific parameters that map to deterministic pre-vetted classes in the Action Registry. Argument validation happens in a sandbox layer, which immunizes the backend from prompt injection and code insertion vulnerabilities.

QA Testing & Validation Metrics

The codebase includes a complete verification test suite containing 48 automated test cases covering profiling accuracy, action registry sanitization, persistence layers, and safety verifications. The success rate is 100% (48/48 tests passed).